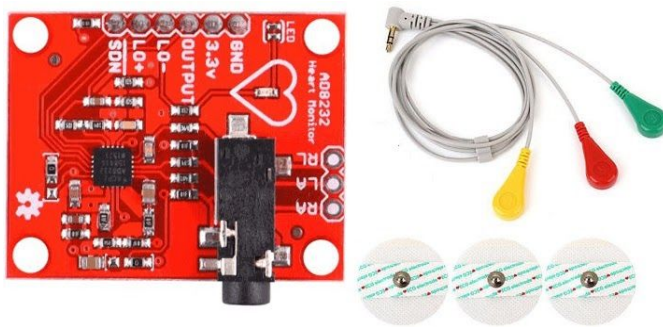


## General description

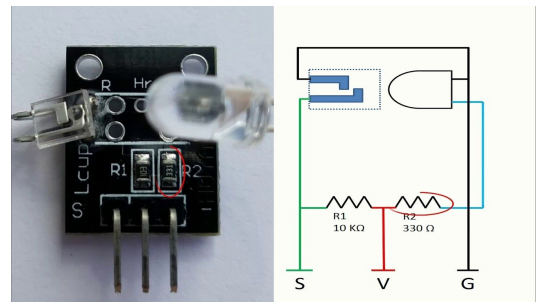
In this project we aim to calculate the heartbeat using an ecg kit (AD8232) and a sensor (KY-039). The KY-039 sensor also gives information about the capacity of the heart to impulse blood to the body extremities.

## Requirements

- arduino uno
- breadboard
- ecg kit AD8232
- KY-039 heartbeat sensor
- lcd display 1602 module
- black or isolate tape
- a computer
- cable usb 2.0 type A/B
- jumper wires



AD8232



KY-039

## Preliminary specification:

Both sensors will be connected to the arduino that will acquire data for the calculations.

## Name of the instrument

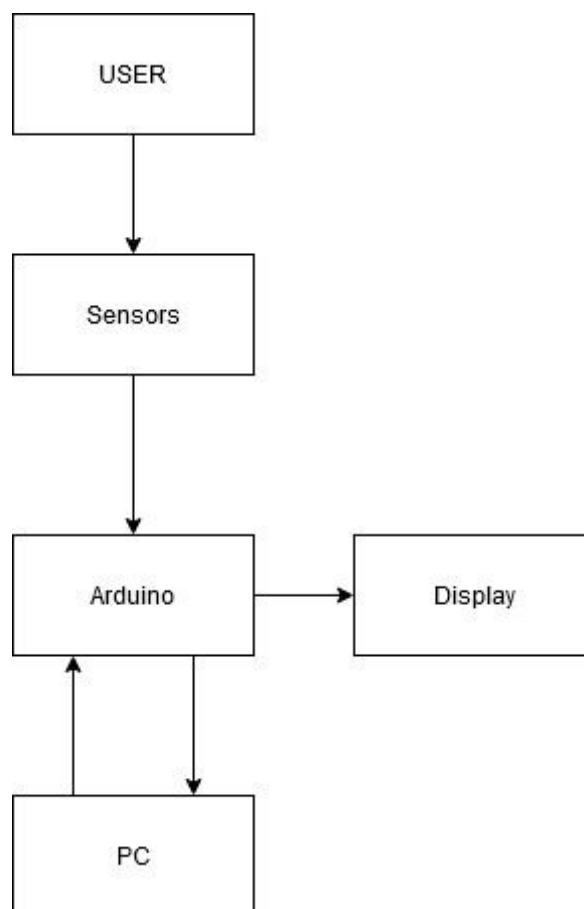
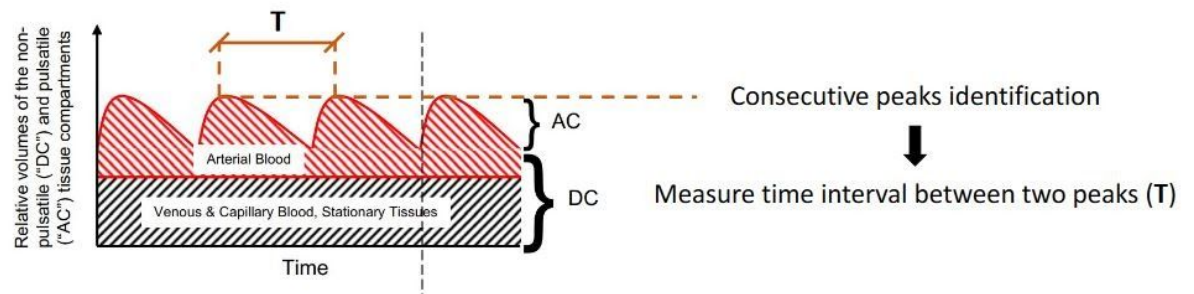
ECG&PULSOMETER

## Function

Our instrument will acquire information from the ECG in simultaneous with the pulsometer. The ECG will give a good information about the BPM of the heart and the pulsometer will give information also about the BPM, that will be compared with the ECG,

but most importantly, it will give information about the capacity of the heart to pump blood to extremities of the body. Both must be used at the same time.

To understand the pulsometer we have to consider that the blood flow is the movement of blood through the vessel where each beat corresponds to a high volume of blood. Since there is a higher volume of blood, there is more infrared light being absorbed and less being received by the sensor, with this change it is possible to determine the heartbeat. The peak amplitude gives information of the pump power.



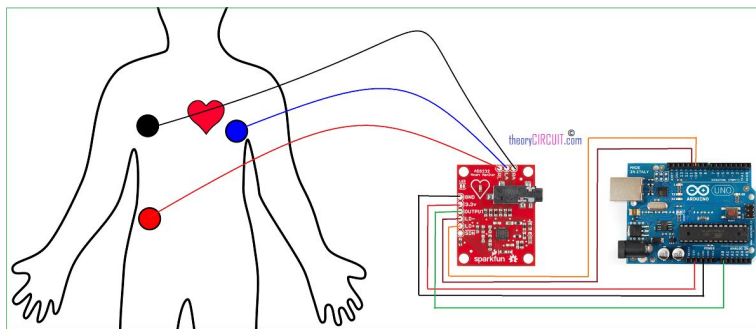
## Details

Our instrument is extremely widespread in the medical-health sector. It can be used at hospitals as well as on emergency vehicles. Analyzing the heart rate with ECG is useful for characterizing heartbeat irregularities (arrhythmias).

The pulsometer gives also information about the pump power of the heart and also its heartbeat and should be putted in the tip of the indicator finger from the left hand.



The instrument will be used by health operators that have the necessity to control the state of the patient. The ECG comes with sticky pads and must be placed according to the figure below: RA (Right Arm), LA (Left Arm), and RL (Right Leg).



## Preconditions

KY-039: The instrument must be placed correctly and we need tape to isolate the finger from the outside light in order to reduce the noise and the patient must not move too much. If in any cases we can't get access to the tap we should consider the noise from ambient light in the data processing.

AD9232: The electrodes must be stucked well to the body to reduce the impedance between skin and electrode essential to reduce the disturbance due to common mode voltage. The operator must be someone with knowledge of how to apply the instruments on the patient.

## Sensors

Generally what arduino provides is an analog value from 0 to 1023.

KY-039: The analog output that we read depends on constant attenuation due to the physiology of the finger plus the attenuation due to the variability of the sanguin flow over time: the higher the output voltage, the less infrared light has been absorbed by the finger.

We see the values on the the serial monitor, the whole process gets timed by the Serial.print() function, which will set a rate for reading and plotting the values.

Observing various simulations on the internet we saw that the printouts change more or less in this range (350 - 400) and has only integer values. As I said before analogical value read by arduino is expressed in the range from 0 to 1028 where 0= 0V and 1028= 5V, with a simple proportion we can say that the output voltage range is included (according to the tests carried out) between 1.50V and 2.00V. The signal will be sampled at approximately 50Hz.

Kit AD8232: Analog Devices' AD8232 heart rate monitoring front-end is an integrated signal conditioning block for ECG. AD8232 adopts a dual-pole high-pass filter to eliminate motion artifacts and electrode half electrode potential, the operational amplifier uses no constraints to build a three-pole low-pass filter, eliminating extra in fact It is designed to extract, amplify and filter small biopotential signals in the presence of noise conditions, such as those created by movement or remote electrode placement.

The signal will be sampled at approximately 1kHz.

According to the Advancement of medical instrumentation (AAMI, 1991), the performance of an electrocardiograph in the frequency band must be between 0.05 and 100Hz, in the case of a simple electrocardiograph to monitor the specifications are less stringent ( 0.5 to 40 Hz). In the sensor used by us for educational use we are "content" with a reduced bandwidth within the last indicated range, this approximation is feasible because the result of the analysis will be only the calculation of the heart rate.

## Signal conditioning

The sensors have already the amplification and hardware filtration implemented and we don't need to condition the analog value.

The output from the KY-039 sensor is analysed with a smoothing filter that takes an average of the 20 last readings from the sensor. It allows us to blunt the signal and eliminate noise due to the 50Hz light frequency in the case of not using isolate tape.

## Data storage

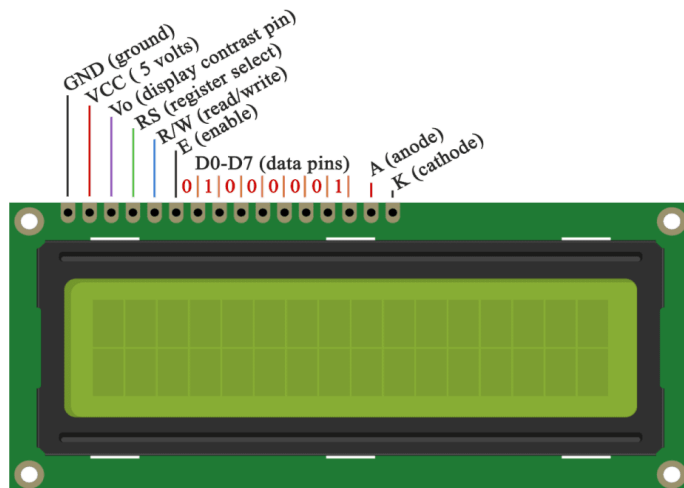
Since the main purpose of the instrument is to determinate the heartbeat and the pump power of the heart continuously there is no reason to storage data, unless we verify that the filtering power of arduino is not enough and we have to do it in matlab.

## Data processing

We will process all data in the arduino for the main reason explained before. After processing the signal, we use the spikes to calculate the BPM. The code will identify the spikes and count how many spikes there are in an interval of time, calculating the BPM. The amplitude of the spikes gives information about the perfusion of blood. If there is small amplitude there is a weak perfusion.

## Display

We will have a lcd display 1602 module that will show the heartbeat from both the ecg and the pulsometer and will also show the pump power of the hearth.



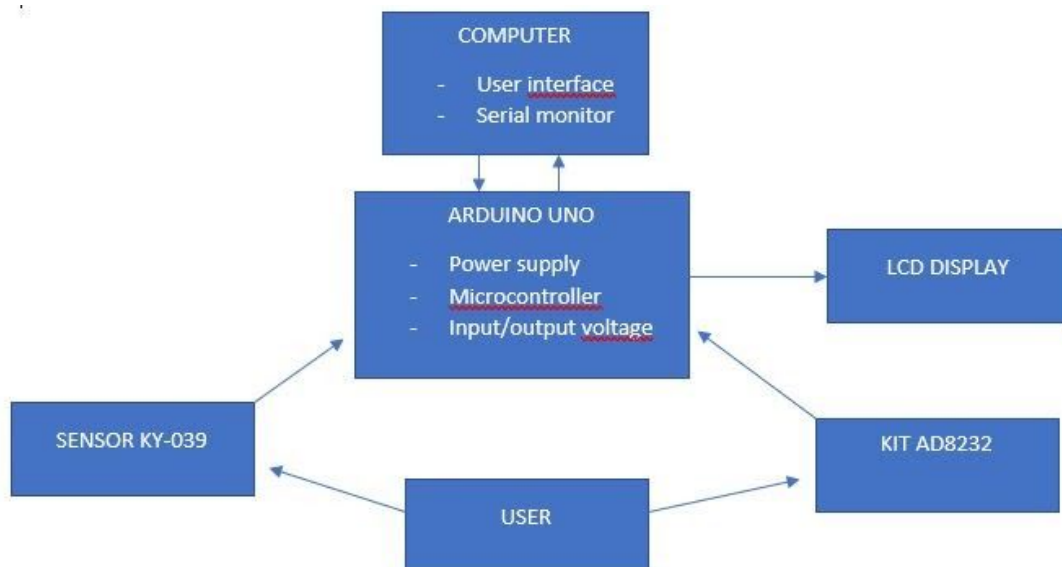
## Power supply

We use the power supply from arduino which is connected to computer with USB 2.0 cable type A/B. We use 5V in output for both sensors.

## Dimensions

The Dimensions is very hard to estimate but we expect not to be bigger than 20cmx20cm.

### Detail block diagram



We use the computer because it's our interface with sensors to write the code and read its results. Arduino is linked to computer with a cable usb 2.0 type A/B which transports binary data and power supply.

Arduino takes analog values from the sensors which will be sampled with a certain frequency decided by us.

At the end of the program, after carefully checking each step, we will print the final values on the display.

### Bill of material (BOM) with cost estimation

3x Sensor KY-039:

[https://www.banggood.com/KY-039-Finger-Detection-Heartbeat-Sensor-Module-Finger-Detect-Measurement-p-1540574.html?rmmds=myorder&cur\\_warehouse=CN](https://www.banggood.com/KY-039-Finger-Detection-Heartbeat-Sensor-Module-Finger-Detect-Measurement-p-1540574.html?rmmds=myorder&cur_warehouse=CN)

Arduino uno:

[https://www.banggood.com/Geekcreit-UNOR3-Basic-Starter-Learning-Kits-No-Battery-Version-Geekcreit-for-Arduino-products-that-work-with-official-Arduino-boards-p-1133595.html?rmmds=myorder&cur\\_warehouse=CN](https://www.banggood.com/Geekcreit-UNOR3-Basic-Starter-Learning-Kits-No-Battery-Version-Geekcreit-for-Arduino-products-that-work-with-official-Arduino-boards-p-1133595.html?rmmds=myorder&cur_warehouse=CN)

ECG kit:

[https://www.banggood.com/DC-3\\_3V-AD8232-ECG-Measurement-Module-Kit-Portable-Heart-Monitor-Biological-p-1136801.html?rmmds=myorder&cur\\_warehouse=CN](https://www.banggood.com/DC-3_3V-AD8232-ECG-Measurement-Module-Kit-Portable-Heart-Monitor-Biological-p-1136801.html?rmmds=myorder&cur_warehouse=CN)

The estimated cost, without considering the computer, is about 25 euros.

### **Task planning**

We write the code to obtain results in the lcd. These results will be the heartbeat from the ECG and pulsometer, that will allow us to see how accurate the sensors are and compared with each other.

The third value will be the amplitude value detected in the pulsometer: if the amplitude value is low, the heart is not pumping blood in an efficient way to the body.

### **Risks identification**

Since there is a low voltage and a low current the risks are almost null.